Southern Nuclear

Regulatory Affairs

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NL-26-0013

ATTN: Document Control Desk
U. S. Nuclear Regulatory Commission
Washington, D. C. 20555-0001

10 CFR 50
10 CFR 54

Edwin I. Hatch Nuclear Plant, Units 1 and 2
Annual Update and Supplement 2 to the Application for Subsequent Renewal of
Operating Licenses, Including Consolidated Implementation Actions Table

Ladies and Gentlemen:

By letter dated May 15, 2025 (ML25135A390), Southern Nuclear Operating Company (SNC) submitted a subsequent license renewal application (SLRA) for Edwin I. Hatch Nuclear Plant Units 1 and 2 (Hatch), Facility Operating License Nos. DPR-57 and NPF-5, to the U.S. Nuclear Regulatory Commission (NRC). By letter dated December 22, 2025, SNC submitted Supplement 1 to the Hatch SLRA (ML25356A512 and ML25356A513). As stated in 10 CFR 54.21(b):

> Each year following submittal of the license renewal application and at least 3 months before scheduled completion of the NRC review, an amendment to the renewal application must be submitted that identifies any change to the CLB [current licensing basis] of the facility that materially affects the contents of the license renewal application, including the FSAR supplement.

In accordance with 10 CFR 54.21(b), SNC has performed an annual review of Hatch CLB changes since the SLRA submittal to determine whether any sections of the SLRA were materially affected by these changes. As a result of this annual review, SNC did not identify any changes to the CLB that materially affect the SLRA.

Enclosure 1 of this letter provides voluntary Supplement 2 markups to SLRA Table 2.2-1 and Section 3.4.2.2.6. Additionally, to support NRC Safety Evaluation Report reference needs and enable an efficient review process, Enclosure 2 of this letter provides a “clean copy” of SLRA Table A-3 with the incorporation of the supplement provided in HNP Supplement 1 letter NL-25-0342 and the SLRA revisions as described in HNP RAI response letter NL-25-0421, Enclosure 1 (ML25356A533 and ML25356A534).

The Enclosure 1 revisions associated with the Hatch SLRA are denoted by ~~strikethrough~~ (deletion) and/or **bold red underline** (insertion) text.

This letter contains no NRC commitments. If you have any questions, please contact Amy Aughtman at 205.992.7006 or agaughtm@southernco.com.

I declare under penalty of perjury that the foregoing is true and correct. Executed on the 26th day of February 2026.

Respectfully submitted,

Digitally signed by JAMIEMCO
Date: 2026.02.26 14:17:30 -06'00'

Jamie M. Coleman
Regulatory Affairs Director

JMC/dsp/cbg

Enclosures: 1) Supplement 2 SLRA Markups
2) SLRA Table A-3 with Supplement Incorporation

Cc: NRC Regional Administrator, Region II
NRC Senior Resident Inspector – Hatch Nuclear Plant
Director, Office of Nuclear Reactor Regulation (NRR)
NRC Project Manager (Safety Review), NRR-DNRL
NRC Project Manager (Environmental Review), NMSS-REFS
NRC Project Manager, NRR-DORL – Hatch Nuclear Plant
Director, Environmental Protection Division – State of Georgia

RTYPE: CHA02.004

Edwin I. Hatch Nuclear Plant Unit 1 and 2
Annual Update and Supplement 2 to the Application for Subsequent Renewal of Operating Licenses, Including Consolidated Implementation Actions Table

Enclosure 1 to NL-26-0013

Supplement 2 SLRA Markups

Revise SLRA to Include Voluntary Clarifications

Affected SLRA Sections: Table 2.2-1 and Section 3.4.2.2.6

SLRA Page Numbers: 2.2-7 and 3.4-8

Description of Change:

SLRA Table 2.2-1 on page 2.2-7 is revised to remove the line item showing that the generator and aux system is in-scope for SLR.

SLRA Section 3.4.2.2.6 on page 3.4-8 is revised to clarify that there are no metallic piping and piping components exposed to raw water or waste water that are susceptible to loss of material due to RIC in the Steam and Power Conversion Systems. Additionally, it is clarified that there is OE indicating that RIC exists for steel components containing raw water within the Engineered Safety Features and Auxiliary Systems.

Table 2.2-1
Plant Level Scoping Report Results

SLRA System Name	HNP System Name	In-scope for SLR	Section
Primary Containment	Drywell to Refueling Basin Expansion Bellows	Y	2.4.1
	Primary Containment	Y	2.4.1
Electrical and I&C Systems			
Electrical Commodities	230Kv Substation	Y	2.5.1
	Analog Transmitter Trip System (ATTS)	Y	2.5.1
	Area Radiation Monitoring System	N	N/A
	Auto-Transformer	N	N/A
	Aux Grounding System	Y	2.5.1
	Breakers	N	N/A
	Cabinets and Panel Boards	Y	2.5.1
	Cable	Y	2.5.1
	Circulating Water Structures Lighting	N	N/A
	Communications Systems (Carrier and Microwave)	N	N/A
	Computer - Nuclear Steam Supply	N	N/A
	Computer - Power Systems Controls	N	N/A
	Control Building; Lighting	N	N/A
	Control Switches	Y	2.5.1
	Coupling Capacitors	N	N/A
	Current Limiting Reactors	N	N/A
	DC Electrical System	Y	2.5.1
	DC Instrument and Control Power	Y	2.5.1
	Drywell Electrical Penetrations	Y	2.5.1
	Electric Boiler	N	N/A
	Electrical Penetrations	Y	2.5.1
	Emergency Response Facilities System	Y	2.5.1
	Feedwater Control System	Y	2.5.1
	Fire Protection	N	N/A
	Fission Products Monitoring System	Y	2.5.1
	Generator and Aux	N	N/A
	~~Generator and Aux~~	~~Y~~	~~2.5.1~~
	Generator Bus	N	N/A
	Heat Trace System	Y	2.5.1
	HP Camera System	N	N/A
	In-Plant Auxiliary Control Panels System	Y	2.5.1
	Instruments	N	N/A
	Isolated Phase Bus	N	N/A
	Lightning Arrestors	N	N/A
	Load Centers	Y	2.5.1

Plant-specific OE examples should be evaluated to determine if the chosen AMP should be augmented even if the thresholds for significance of aging effect or frequency of occurrence of aging effect have not been exceeded. For example, during a 10 year search of plant-specific OE, two instances of a 360º 30 percent wall loss occurred at copper alloy to steel joints. Neither the significance of the aging effect nor the frequency of occurrence of aging effect threshold has been exceeded. Nevertheless, the OE should be evaluated to determine if the AMP that is proposed to manage the aging effect is sufficient (e.g., method of inspection, frequency of inspection, number of inspections) to provide reasonable assurance that the CLB intended functions of the component will be met throughout the subsequent period of extended operation. While recurring internal corrosion is not as likely in environments other than raw water and waste water (e.g., treated water), the aging effect should be addressed in a similar manner.

There are no metallic piping or piping components exposed to raw water or waste water that are susceptible to loss of material due to recurring internal corrosion in the Steam and Power Conversion Systems. However, HNP OE over the past 10 years indicates that recurring internal corrosion **does** exist~~s~~ for steel components containing raw water from the Altamaha River **within the Engineered Safety Features and Auxiliary Systems.** ~~; t~~**T**herefore, recurring internal corrosion is an applicable aging effect at HNP.

Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components (B.2.3.24) AMP is a new AMP and will address recurring internal corrosion by having measures in place to inspect additional samples for recurring degradation to ensure that corrective actions are appropriately addressing the associated causes. For ongoing degradation mechanisms (e.g., MIC and erosion) or recurring loss of material due to internal corrosion, the frequency and extent of wall thickness inspections will be increased commensurate with the significance of the degradation. If the cause of the aging effect for each applicable material and environment is not corrected by repair or replacement for all components constructed of the same material and exposed to the same environment, additional inspections will be conducted if one of the inspections does not meet acceptance criteria. The number of inspections will be increased in accordance with the CAP; however, no fewer than five additional inspections will be conducted for each inspection that does not meet acceptance criteria, or 20 percent of each applicable material, environment, and aging effect combination are inspected, whichever is less.

Therefore, there is no need to augment the Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components (B.2.3.24) AMP because it already includes specific measures to detect and manage recurring internal corrosion.

3.4.2.2.7 Cracking Due to Stress Corrosion Cracking in Aluminum Alloys

The SCC is a form of environmentally assisted cracking which is known to occur in high and moderate strength aluminum alloys. The three conditions necessary for SCC to occur in a component are a sustained tensile stress, aggressive environment, and material with a susceptible microstructure. Cracking due to SCC can be mitigated by eliminating one of the three necessary conditions. For the purposes of SLR, acceptance criteria for this further evaluation are provided for

Edwin I. Hatch Nuclear Plant Unit 1 and 2

Annual Update and Supplement 2 to the Application for Subsequent Renewal of Operating Licenses, Including Consolidated Implementation Actions Table

Enclosure 2 to NL-26-0013

SLRA Table A-3 with Supplement Incorporation

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
1	Environmental Qualification of Electric Equipment (A.2.1.1)	X.E1	Continue the existing Environmental Qualification of Electric Equipment AMP, including enhancements to: a) Include monitoring or inspection of certain environmental conditions, including ALEs, or equipment parameters to verify that the equipment is within the bounds of its qualification basis, or as a means of modifying the qualified life. b) Visually inspect accessible, passive EQ equipment at least once every 10 years with the first periodic visual inspection being performed prior to the SPEO. c) Document within the visual inspections that accessible passive EQ equipment is free from unacceptable surface abnormalities that may indicate age degradation. d) Enhance procedures to evaluate and take appropriate corrective actions, which may include changes to qualified life, when an unexpected ALE or condition is identified during operational or maintenance activities that affect the qualification of electrical equipment.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the 10-year interval inspections no earlier than 10 years prior to the SPEO.
2	Fatigue Monitoring (A.2.1.2)	X.M1	Continue the existing Fatigue Monitoring AMP, including enhancements to: a) Update plant procedure(s) to provide procedural direction to require periodic validation of chemistry parameters that are used as inputs to determine F_{en} factors. b) Update the Fatigue Monitoring AMP governing procedure to identify and require monitoring of the 80-year plant design cycles, or projected cycles that are utilized as inputs to component CUF_{en} calculations, as applicable.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			c) Update plant procedures to identify the corrective action options to take if the values assumed for fatigue parameters are approached, transient severities exceed the design or assumed severities, transient counts exceed the design or assumed quantities, transient definitions have changed, unanticipated new fatigue loading events are discovered, or the geometries of components are modified. d) The procedure governing the Fatigue Monitoring AMP will be enhanced to specify that for CUFen analyses, scope expansion includes consideration of other locations with the highest expected CUFen values.	
3	Neutron Fluence Monitoring (A.2.1.3)	X.M2	Implement the new Neutron Fluence Monitoring AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
4	ASME Section XI Inservice Inspection, Subsections IWB, IWC, and IWD (A.2.2.1)	XI.M1	Continue the existing ASME Section XI Inservice Inspection, Subsections IWB, IWC, and IWD AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
5	Water Chemistry (A.2.2.2)	XI.M2	Continue the existing Water Chemistry AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
6	Reactor Head Closure Stud Bolting (A.2.2.3)	XI.M3	Implement the new Reactor Head Closure Stud Bolting AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
7	BWR Vessel ID Attachment Welds (A.2.2.4)	XI.M4	Continue the existing BWR Vessel ID Attachment Welds AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
8	BWR Stress Corrosion Cracking (A.2.2.5)	XI.M7	Continue the existing BWR Stress Corrosion Cracking AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
9	BWR Penetrations (A.2.2.6)	XI.M8	Continue the existing BWR Penetrations AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
10	BWR Vessel Internals (A.2.2.7)	XI.M9	Continue the existing BWR Vessel Internals AMP, including enhancements to: a) Add BWRVIP-19-A which provides guidelines for repair design criteria into the Vessel and Internals program and add to the Vessel and Internals Program – Reactor Vessel Internals Bases Document. b) Include implementation of BWRVIPs -26-A, -41-R4-A, -47-A, and -183-A, or any future NRC-approved revisions of these documents, as indicated in Appendix B of BWRVIP-315-A. c) A revision to BWRVIP-315-A, “Reactor Internals Aging Management Evaluation for Extended Operations” was published in April 2024 (ML24191A266). A proprietary and NP version was	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			forwarded to NRC on 6/18/24 (ML24191A244). HNP has initiated a tracking item for implementing this new version of BWRVIP-315 as well as approved revisions to the above BWRVIP guidance documents into the program. d) Add BWRVIP-234-A to the Vessel and Internals program and add to the Vessel and Internals Program – Reactor Vessel Internals Bases Document.	
11	Thermal Aging Embrittlement of Cast Austenitic Stainless Steel (A.2.2.8)	XI.M12	Implement the new Thermal Aging Embrittlement of Cast Austenitic Stainless Steel AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
12	Flow-Accelerated Corrosion (A.2.2.9)	XI.M17	Continue the existing Flow-Accelerated Corrosion AMP, including enhancements to: a) Reassess piping systems excluded from wall thickness monitoring due to operation less than 2 percent of plant operating time (as allowed by NSAC-202L-R4) to ensure the exclusion remains valid and applicable for operation beyond 60 years. b) Formalize a separate erosion susceptibility evaluation (ESE) that will include all components determined to be susceptible to wall loss due to erosion through OE and industry guidance. c) Revise or provide procedure(s) for measuring wall thickness due to erosion. Wall thickness should be trended to adjust the	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			monitoring frequency and to predict the remaining service life of the component for scheduling repairs or replacements. d) Revise or provide procedure(s) to evaluate inspection results to determine if assumptions in the extent-of-condition review remain valid. If degradation is associated with infrequent operational alignments, such as surveillance or pump starts/stops, then trending activities should consider the number or duration of these occurrences. e) Revise or provide procedure(s) to perform periodic wall thickness measurements of replacement components until the effectiveness of corrective actions have been confirmed. f) Revise or provide procedure(s) to provide guidance consistent with the erosion service life safety factor from EPRI 3002023786 for locations with known erosion mechanisms. Changes that recommend an increase in safety factor to 2.0 will be documented. g) System exclusion for cavitation screening will be based on less than 100 hours of operation per year.	
13	Bolting Integrity (A.2.2.10)	XI.M18	Continue the existing Bolting Integrity AMP, including enhancements to: a) Create a procurement document to ensure that the maximum yield strength of replacement or newly- procured pressure-retaining bolting material will be limited to an actual yield strength less than 150 ksi.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			b) Clarify that MoS2 and other lubricants containing sulfur will be prohibited from use on pressure-retaining closure bolting. c) Ensure that closure bolting where leakage is difficult to detect (e.g., piping systems that are submerged or that contain compressed air or gas). The acceptance criteria for the alternative means of testing will be no indication of leakage from the bolted connections. Inspections will be performed to ensure that a representative sample of the population (defined as the same material and environment combination) of bolt heads and threads is accessed over each 10-year period of the SPEO. The representative sample will be 20 percent of the population (up to a maximum of 19 per Unit). d) Ensure non-ASME Section XI high-strength closure bolting (with actual yield strengths greater than or equal to 150 ksi), and bolting for which yield strength is unknown, will be monitored for surface and subsurface discontinuities indicative of cracking. e) Ensure that bolted joints in areas that are accessible during normal operation are visually inspected once per refueling cycle. f) Ensure that bolted joints not readily visible during plant operations and refueling outages will be visually inspected when they are made accessible and at such intervals that would provide reasonable assurance the components' intended functions are maintained. g) Ensure that closure bolting greater than two inches in diameter (regardless of code classification) with actual yield strength greater than or equal to 150 ksi and closure bolting for which yield strength is unknown will require volumetric examination in accordance to	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			that of ASME Code Section XI, Table IWB-2500-1, Examination Category B-G-1. h) Project, where practical, identified degradation until the next scheduled inspection. Results will be evaluated against acceptance criteria to confirm that the timing of subsequent inspections will maintain the components' intended functions throughout the SPEO based on the projected rate of degradation. For sampling-based inspections, results will be evaluated against acceptance criteria to confirm that the sampling bases will maintain the components' intended functions throughout the SPEO based on the projected rate and extent of degradation. Adverse results will be evaluated to determine if an increased sample size or inspection frequency is required. i) Include the guidance for corrective action in response to joint leakage (i.e., sample expansion and additional inspections if inspection results do not meet acceptance criteria).	
14	Open-Cycle Cooling Water System (A.2.2.11)	XI.M20	Continue the existing Open-Cycle Cooling Water AMP, including enhancements to: a) Update the piping inspection procedures to monitor for internal cracking in copper alloys with greater than 15 percent zinc. b) Update the service water systems heat exchanger testing procedures to require the final heat exchanger testing frequency to be at least once every five years. c) Ensure Non-ASME code tests and inspections follow site procedures that include requirements for items such as lighting,	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			distance, offset, surface coverage, presence of protective coatings, and cleaning processes. d) Clarify in heat exchanger testing and inspection procedures that inspection results are trended to evaluate the adequacy of surveillance frequencies so that proper function is maintained between surveillances. e) Ensure the service water program and heat exchanger inspection procedures prompt an evaluation of the heat transfer capability of the safety-related, raw water supplied heat exchangers when fouling is identified. f) Ensure service water program and inspection procedures include trending of wall thickness measurements at locations susceptible to ongoing degradation due to specific aging mechanisms (e.g., MIC). Ensure the monitoring frequency and number of inspection locations is adjusted based on the trending. g) Update the service water program procedure and inspection and testing procedures to clarify that if fouling is identified, the overall effect is evaluated for flow blockage, loss of material, reduction in heat transfer, and chemical treatment effectiveness. For ongoing degradation mechanisms (e.g., MIC) or loss of material due to recurring internal corrosion, the frequency and extent of wall thickness inspections will be increased commensurate with the significance of the degradation. If the cause of the aging effect for each applicable material and environment is not corrected by repair or replacement for all components constructed of the same material and exposed to the same environment, additional inspections are conducted if one of the inspections does not	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			meet acceptance criteria. The service water program procedure will also be updated to state that the number of inspections will be increased in accordance with the CAP; however, no fewer than five additional inspections are conducted for each inspection that did not meet acceptance criteria, or 20 percent of each applicable material, environment, and aging effect combination is inspected, whichever is less. Regardless of which unit the condition is identified, the additional number of inspections will be performed on Unit 1 and Unit 2 where the same material, environment, and aging effect combination exists (each unit will perform a minimum of five additional inspections or 20 percent of each applicable material, environment, and aging effect combination, whichever is less).	
15	Closed Treated Water Systems (A.2.2.12)	XI.M21A	Continue the existing Closed Treated Water Systems AMP, including enhancements to: a) Update implementing procedure(s) or create new procedure(s) to include evaluation of the visual appearance of surfaces for evidence of loss of material. The results of surface or volumetric examinations will be evaluated for surface discontinuities indicative of cracking. The heat transfer capability of heat exchanger surfaces will be evaluated by either visual inspections to determine surface cleanliness, or functional testing to verify that design heat removal rates are maintained. b) Update implementing procedure(s) to include monitoring frequencies that are in accordance with the latest version of the	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			EPRI Closed Cooling Water Chemistry Guideline for non-glycol systems. c) Update implementing procedure(s) or create new procedure(s) to include visual inspection of surfaces exposed to the closed treated water environment for evidence of loss of material, cracking, or fouling (of heat transfer surfaces) whenever the system boundary is opened. At a minimum, in each 10-year period during the SPEO, a representative sample (20 percent of the population, up to a maximum of 25 components) of piping and components will be inspected using techniques capable of detecting loss of material, cracking, and fouling, as appropriate. The 20 percent minimum is surface area inspected unless the component is measured in linear feet, such as piping. In that case, any combination of 1-foot length sections and components can be used to meet the recommended extent of 25 inspections. The representative sample will be selected based on likelihood of corrosion or cracking. Inspections will be conducted in accordance with applicable ASME code requirements. If there are no ASME code requirements, inspections will be conducted in accordance with site procedures, which will include requirements for items such as lighting, distance, offset, surface coverage, presence of protective coatings, and cleaning processes. d) Update implementing procedure(s) or create new procedure(s) to include acceptance criteria for the results of visual inspection of surfaces exposed to the closed treated water environment. Any detectable loss of material, cracking, or fouling (of heat transfer surfaces) will be evaluated in the CAP.	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			e) Update implementing procedure(s) or create new procedure(s) to include corrective actions if the results of visual inspection of surfaces exposed to the closed treated water environment do not meet acceptance criteria. If fouling of heat transfer surfaces is identified, the overall effect will be evaluated for reduction of heat transfer, flow blockage, and loss of material. If the cause of the aging effect for each applicable material and environment is not corrected by repair or replacement for all components constructed of the same material and exposed to the same environment, additional inspections are conducted if one of the inspections does not meet acceptance criteria. Corrective actions will include additional inspections. The number of increased inspections will be determined in accordance with the CAP; however, there will be no fewer than five additional inspections for each inspection that did not meet acceptance criteria, or 20 percent of each applicable material, environment, and aging affect inspected, whichever is less. If subsequent inspections do not meet acceptance criteria, an extent of condition and extent of cause analysis will be conducted to determine the further extent of condition. Additional samples will be inspected for any recurring degradation to ensure corrective actions appropriately address the associated causes. The additional inspections include inspections at all of the HNP units that have the same material, environment, and aging effect combination. The additional inspections will be completed within the interval (e.g., refueling outage interval, 10-year inspection interval) in which the original inspection was conducted.	
16	Inspection of Overhead Heavy	XI.M23	Continue the existing Inspection of Overhead Heavy Load and Light Load (Related to Refueling) Handling Systems AMP, including enhancements to:	No later than the last refueling outage prior to the SPEO, or no

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
	Load and Light Load (Related to Refueling) Handling Systems (A.2.2.13)		a) Update procedures to state the visual inspection frequencies required by the 1976 version of ASME B30.2. A crane that is used in infrequent service, which has been idle for a period of one year or more, shall be inspected by a designated person and documented before being placed in service in accordance with the requirements listed in ASME B30.2 paragraph 2-2.1.3 (i.e., periodic inspection). b) Update procedures to specifically require visual inspectors to be VT qualified in accordance with plant-specific procedures and processes. c) Update procedures to state that repairs made to NUREG-0612 load handling systems are performed as specified in the 1976 version of ASME B30.2. d) Update procedures to state that any visual indication of loss of material, deformation, or cracking, and any visual sign of loss of bolting preload is evaluated as required by ASME B30.2 or other applicable industry standard in the ASME B30 series. e) Update procedures to state that any unacceptable indication of loss of material will be evaluated by engineering. When appropriate, engineering evaluations will be based upon the design code of record. If warranted, additional inspections will be performed. Any significant degradation of components is noted and corrective actions will be implemented in accordance with the CAP. f) Update procedures to state that the acceptance criteria for steel structures and connections guidelines from the Structural Monitoring Program for the maintenance shall be used.	later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
17	Compressed Air Monitoring (A.2.2.14)	XI.M24	Implement the new Compressed Air Monitoring AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
18	Fire Protection (A.2.2.15)	XI.M26	Continue the existing Fire Protection AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
19	Fire Water System (A.2.2.16)	XI.M27	Continue the existing Fire Water System AMP, including enhancements to: a) Caulking or sealant at the tanks' foundation will be inspected annually. Visual inspections are supplemented by conducting physical manipulation to detect degradation. Nonpliable, cracked, or missing sealant and caulking is unacceptable. When degraded sealant or caulking is detected, an evaluation is conducted to determine the need to conduct follow-up examinations of the tank's surface. b) Update inspection procedures such that in each 5-year interval of the SPEO, 20 percent of the length of piping segments that cannot be drained or piping segments that allow water to collect is subject to volumetric wall thickness inspections. Measurement points are obtained to the extent that each	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the pre-SPEO tests and inspections during the 10 years prior to the SPEO for the fire water storage tanks.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			potential degraded condition can be identified (e.g., general corrosion, MIC). The 20 percent of piping that is inspected in each 5-year interval is in different locations than previously inspected piping. c) In each 5-year interval, beginning five years prior to the SPEO, either conduct a flow test or flush sufficient to detect potential flow blockage, or conduct a visual inspection of 100 percent of the internal surface of piping segments that cannot be drained or piping segments that allow water to collect. d) Ensure a flow test will be conducted in the 5 year period prior to the SPEO at the hydraulically most remote hose connection of each building to verify the water supply. Where a flow test of the hydraulically most remote hose connection is not practical, engineering will be consulted for the appropriate location for the test. Subsequent tests at the most remote hose connection during the SPEO will be conducted every 5 years on a representative sample of 20 percent of the population (defined as components having the same material and environment combination) or a maximum of 25 per population at each unit. e) Perform volumetric inspections or low-frequency electromagnetic testing of all fire water storage tanks walls if there are signs of degradation. Tank bottoms will be tested for loss of material on the underside by use of ultrasonic testing where there is evidence of pitting or corrosion. An acceptable alternative to ultrasonic testing is the removal, visual inspection, and replacement of random floor coupons.	For other components with pre-SPEO inspections and tests, implement during the 5 years prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			f) Ensure when there are signs or age-related degradation detected in the vicinity of welds, the fire water storage tanks shall be vacuum-box tested at bottom seams in accordance with test procedures found in NFPA 22. g) Update fire water storage tank inspections to have the bottom surfaces inspected. Specifically, for each 10-year period starting 10 years before the SPEO, low-frequency electromagnetic testing or volumetric inspection of the tank bottom will be performed from the inside surface of the tanks. Any regions below nominal plate thickness (in excess of plate manufacturing tolerance) will have a follow-up ultrasonic thickness reading. If there are areas of significant loss of material that could impact the pressure boundary function, future ultrasonic thickness measurements and trending will be performed. h) For main drain tests, ensure when there is a 10 percent reduction in full flow pressure when compared to the original acceptance test or previously performed tests, the cause of the reduction shall be identified and corrected if necessary. i) An internal inspection of piping and branch line conditions shall be conducted at least every 5 years by opening a flushing connection at the end of one main and by removing a sprinkler toward the end of one branch line for the purpose of inspecting for the presence of foreign organic and inorganic material. 1. Update to permit alternative nondestructive examination methods that can ensure that flow blockage will not occur.	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			2. Include inspections to discuss tubercules or slime, if found, shall be tested for indications of microbiologically influenced corrosion (MIC). 3. State if the presence of sufficient foreign organic or inorganic material is found to obstruct pipe or sprinklers, an obstruction investigation shall be conducted as described in NFPA 25 Section 14.3. 4. State that inspection of a cross main is not required where the system does not have a means of inspection. 5. State that if loose deposits are identified in the piping, and the evaluation determines that the deposits must be removed, then the piping is required to be flushed repeatedly, in accordance with NFPA 25 Annex D.5, until it is determined that either no deposits are left or that the remaining deposits pose no blockage threat. Areas where excessive deposits are found will undergo more thorough volumetric wall testing to ensure minimum wall thickness is met. j) Conduct an obstruction investigation for system or yard main piping wherever any of the following conditions exist: 1. The discharge of obstructive material during routine water tests 2. Foreign materials in fire pumps, in dry pipe valves, or in check valves	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			3. Foreign material in water during drain tests or plugging of inspector's test connection(s) 4. Plugged sprinklers 5. Plugged piping in sprinkler systems dismantled during building alterations 6. Pinhole leaks 7. A 50 percent increase in the time it takes water to travel to the inspector's test connection from the time the valve trips during a full flow trip test of a dry pipe sprinkler system when compared to the original system acceptance test. If any of the above obstruction conditions are present, HNP will initiate a CR and will be investigated and resolved through the CAP. k) Clarify to include monitoring and trending of data to confirm that components will maintain their intended functions throughout the SPEO based on projected rate and extent of degradation if any degradation is identified. l) Clarify to measure wall thickness to compare the wall thickness to the minimum design if there is degradation identified. Wall thicknesses less than the minimum design will be entered into the CAP process for engineering evaluation. m) Demonstrate that no loose fouling products exist in the systems that could cause flow blockage in the sprinklers or deluge nozzles. n) When all the flow tests at the most hydraulically remote hose connections of each building conducted no earlier than 5 years	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			prior to the subsequent period of extended operation meet the design pressure at the required flow acceptance criteria, subsequent tests of the most remote hose connections may be conducted on a representative sample of 20 percent of the population (defined as components having the same material and environment combination) or a maximum of 25 per population at each unit. o) Apply training and personnel qualification requirements, corrective actions, and acceptance criteria from Section A.2.2.28 to the Fire Water Storage Tank Coatings/Linings inspections. p) If a flow test or main drain test does not meet the acceptance criteria due to current or projected degradation, additional tests are conducted in locations that could be susceptible to the same obstruction material and were not initially tested. The additional inspections are completed during the interval (i.e., 5 years, annual) in which the original test was conducted. If subsequent inspections do not meet acceptance criteria, an extent of condition and extent of cause will be conducted to determine further extent of inspection. q) Update inspection procedure to ensure that a safety factor of at least 2.0 will be used for determinations of erosion mechanism re-inspection intervals for SSCs deemed susceptible to erosion by susceptibility analysis.	
20	Outdoor and Large Atmospheric Metallic	XI.M29	Continue the existing Outdoor and Large Atmospheric Metallic Storage Tanks AMP, including enhancements to:	No later than the last refueling outage prior to the SPEO, or

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
	Storage Tanks (A.2.2.17)		a) Update the implementation procedure title to reflect that this AMP requires more than visual examinations on the CSTs. This procedure references other examination types used in CST examinations that satisfy the requirements of the current LR. b) Update section 2.0 of the AMP implementation procedure to reflect the applicability and frequency of the inspections outlined in the AMP basis document. c) Update the AMP implementation procedure examination checklist to include inspection areas exposed to all tank environments (i.e., ensure test sample group includes tank wall areas exposed to air/condensation and water). d) Perform an internal volumetric inspection of the tank bottom for loss of material and cracking. The frequency is “each 10-year period starting 10 years before the SPEO." e) Enhance the AMP implementation procedure to include the following: when degraded conditions are identified, acceptability of the condition is projected to the next scheduled inspection or in the case of one-time inspections, follow-up inspections are scheduled. f) Enhance the AMP implementation procedure to include the following: results are evaluated against acceptance criteria to confirm or adjust timing of subsequent inspections, or in the case of one-time inspections, schedule follow-up inspections. g) Enhance the examination section of the AMP implementation procedure to include the coated areas for	no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the pre-SPEO inspections no earlier than 10 years prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			examination and to add a specific examination task to inspect the condition of coatings, sealants, and caulking along with physical manipulation of sealants and caulking. h) Enhance the sample set expansion requirements of the AMP implementation procedure to state the number of increased inspections is determined in accordance with the site's CAP. However, for other sampling-based inspections (e.g., 20 percent, 25 locations) the smaller of five additional inspections or 20 percent of the inspection population will be conducted. If subsequent inspections do not meet acceptance criteria, an extent of condition and extent of cause will be conducted to determine the further extent of inspection.	
21	Fuel Oil Chemistry (A.2.2.18)	XI.M30	Continue the existing Fuel Oil Chemistry AMP, including enhancements to: a) Create a new implementing procedure for the fire pump diesel storage tanks cleaning and inspections, which has the following. 1. Steps for cleaning and volumetric examinations for signs of loss of material every 10 years. If there are signs of loss of material of tank wall thickness, include steps for repairs or additional evaluations. 2. Identified degradation is projected until the next scheduled inspection, where practical. 3. Results are evaluated against acceptance criteria to confirm that the components will maintain the intended functions based on the projected rate of degradation throughout the SPEO.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			4. Thickness measurements are evaluated against the design thickness and corrosion allowance. Any degradation identified during volumetric examinations is reported and evaluated using the CAP. b) Ensure that the following inspections and examinations are included in the FOST inspections: 1. Steps for repairs or volumetric inspections of the tank wall thickness if there are any signs of corrosion during the visual inspections every 10 years. 2. Identified degradation is projected until the next scheduled inspection, where practical. 3. Results are evaluated against the acceptance criteria to confirm that the components will maintain the intended functions based on the projected rate of degradation throughout the SPEO. 4. Thickness measurements are evaluated against the design thickness and corrosion allowance. Any degradation identified during volumetric examinations is reported and evaluated using the CAP. c) Ensure that the following inspections and examinations are included in the diesel generator day tank inspections: 1. Steps for cleaning and visual inspections for signs of corrosion of the diesel generator day tanks every 10 years. Include steps for repairs or additional volumetric	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			examinations to verify the tank wall thickness if there are signs of corrosion. 2. Identified degradation is projected until the next scheduled inspection, where practical. 3. Results are evaluated against the acceptance criteria to confirm that the components will maintain the intended functions based on the projected rate of degradation throughout the SPEO. 4. Any visual degradation of the diesel generator day tanks is reported and evaluated by generating a CR. 5. Thickness measurements are evaluated against the design thickness and corrosion allowance. Any degradation during the visual inspections or volumetric inspections is reported and evaluated using the CAP. d) Ensure that the following is updated in the fuel oil chemistry inspections: 1. Testing, monitoring, and trending frequency for biological activity will be updated to at least quarterly.	
22	Reactor Vessel Material Surveillance (A.2.2.19)	XI.M31	Continue the existing Reactor Vessel Material Surveillance AMP, including enhancement to: a) Implement BWRVIP-321 Revision 1-A to maintain compliance with 10 CFR 50, Appendix H. during the SPEO.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
23	One-Time Inspection (A.2.2.20)	XI.M32	Implement the new One-Time Inspection AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the one-time inspections no earlier than 10 years prior to the SPEO.
24	Selective Leaching (A.2.2.21)	XI.M33	Implement the new Selective Leaching AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the one-time and 10-year interval inspections no earlier than 10 years prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
25	ASME Code Class 1 Small-Bore Piping (A.2.2.22)	XI.M35	Implement the new ASME Code Class 1 Small-Bore Piping AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Start the one-time and 10-year interval inspections no earlier than 6 years prior to the SPEO.
26	External Surfaces Monitoring of Mechanical Components (A.2.2.23)	XI.M36	Implement the new External Surfaces Monitoring of Mechanical Components AMP.	No later than the last refueling outage prior to the SPEO or no later than 6 months prior to the SPEO, i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
27	Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components (A.2.2.24)	XI.M38	Implement the new Inspection of Internal Surfaces in Miscellaneous Piping and Ducting Components AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
28	Lubricating Oil Analysis (A.2.2.25)	XI.M39	Continue the existing Lubricating Oil Analysis AMP, including enhancement to: a) Update the lubricating oil analysis procedures to clarify that phase-separated water in any amount is not acceptable.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
29	Monitoring of Neutron-Absorbing Materials Other Than Boraflex (A.2.2.26)	XI.M40	Continue the existing Monitoring of Neutron-Absorbing Materials Other Than Boraflex AMP, including enhancements to: a) Update the surveillance sample removal and installation procedure to assure EPRI good practices are being used to maintain acceptable chemistry parameters and the corresponding coupon inspection PM tasks specifically document any aluminum oxide layer degradation on the coupon.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
30	Buried and Underground Piping and Tanks (A.2.2.27)	XI.M41	Continue the existing Buried and Underground Piping and Tanks AMP, including enhancements to: a) Update the Excavation & Earthwork Quality Control procedure to state that new and replacement backfill shall meet the requirements of NACE SP-0169-2007 Section 5.2.3 or NACE RP-0285-2002 Section 3.6. Backfill that is located within 6 inches of the component that meets ASTM D 448-08 size number 67 (size number 10 for polymeric materials) is considered to meet the objectives of NACE SP0169-1 2007 and NACE RP0285-2002. For stainless steel, backfill limits apply only if the component is coated. The use of controlled low-strength materials (flowable	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the 10-year interval inspections no earlier than 10 years prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			backfill) is also acceptable to meet the objectives of NACE SP0169-2007. b) Update the BUPT implementing procedure to perform soil testing during excavations for inspections. Additionally, include a requirement to perform an evaluation at least every five years during the SPEO to ensure the soil samples taken during that period are representative of the vicinity in which in-scope components are buried. c) Install cathodic protection on the diesel fuel oil storage tanks prior to the SPEO. The cathodic protection system installed on the diesel fuel oil storage tanks will be in accordance with NACE SP0169-2007 or NACE RP0285-2002. d) Update procedures and/or specifications to require that newly installed, buried stainless steel piping is coated in accordance with Table 1 of National Association of Corrosion Engineers (NACE) SP0169-2007 or Section 3.4 of NACE RP0285-2002. e) Update the BUPT implementing procedure to monitor for crevice corrosion and MIC for steel (including ductile and gray cast iron), and stainless steel components. f) Update the BUPT implementing procedure to clarify that inspections for cracking due to SCC for stainless steel and steel (in a carbonate-bicarbonate environment) utilize a method that has been determined to be capable of detecting cracking. Coatings that: (a) are intact, well-adhered, and otherwise sound for the remaining inspection period; and (b) exhibit small blisters that are few in number and completely surrounded by sound coating bonded to the substrate do not have to be removed. Inspections for	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			cracking are conducted to assess the impact of cracks on the pressure boundary function of the component. g) Update fuel oil storage tank inspection and cleaning tasks to align with a 10-year frequency. h) Update the BUPT implementing procedure to specify that visual inspections are supplemented with surface and/or volumetric nondestructive testing if evidence of wall loss beyond minor surface scale is observed. i) Update the BUPT implementing procedure to include the following: Where practical, identified degradation (e.g., coating condition) is projected until the next scheduled inspection occurs. Results are evaluated against acceptance criteria to confirm that the sampling bases (e.g., selection, size, frequency) will maintain the components' intended functions throughout the SPEO based on the projected rate and extent of degradation. For any installed cathodic protection system, potential difference and current measurements are trended to identify changes in the effectiveness of the system and/or coatings. j) Update the BUPT implementing procedure to include the following acceptance criteria: 1. For coated piping or tanks, there is either no evidence of coating degradation, or the type and extent of coating degradation is evaluated as being insignificant by (1) an individual who has a NACE Coating Inspector Program Level 2 or 3 inspector qualification; (2) an individual who has completed the EPRI Comprehensive Coatings Course and completed the EPRI Buried Pipe Condition Assessment and	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			Repair Training Computer Based Training Course; or (3) a coatings specialist qualified in accordance with an ASTM standard endorsed in RG 1.54, Revision 2, "Service Level I, II, and III Protective Coatings Applied to Nuclear Power Plants." 2. The measured wall thickness projected to the end of the SPEO meets minimum wall thickness requirements. 3. Indications of cracking in metallic pipe are managed in accordance with the "corrective actions" program element. 4. Backfill is acceptable if the inspections do not reveal evidence that the backfill caused damage to the component's coatings or the surface of the component (if not coated). 5. Cracks in cementitious backfill that could admit groundwater to the surface of the component are not acceptable. k) Update the BUPT implementing procedure to include the following corrective actions: 1. Where damage to the coating has been evaluated as being significant and the damage was caused by nonconforming backfill, an extent of condition evaluation is conducted to determine the extent of degraded backfill in the vicinity of the observed damage. 2. If coated or uncoated metallic piping or tanks show evidence of corrosion, the remaining wall thickness in the affected	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			area is determined to ensure that the minimum wall thickness is maintained. This may include different values for large area minimum wall thickness and local area wall thickness. If the wall thickness extrapolated to the end of the SPEO meets the minimum wall thickness requirements, the recommendations for expansion of sample size below do not apply. 3. When the coatings, backfill, or condition of exposed piping does not meet the acceptance criteria, the degraded condition is repaired or the affected component is replaced. In addition, when the depth or extent of degradation of the base metal could have resulted in a loss of pressure boundary function when the loss of material is extrapolated to the end of the SPEO, an expansion of sample size is conducted. The number of inspections within the affected piping categories is doubled or increased by five, whichever is smaller. If the acceptance criteria are not met in any of the expanded samples, an analysis is conducted to determine the extent of condition and extent of cause. The number of follow-on inspections is determined based on the extent of condition and extent of cause. 4. The timing of the additional examinations is based on the severity of the degradation identified and is commensurate with the consequences of a leak or loss of function. However, in all cases, the expanded sample inspection is completed within the 10-year interval during which the original inspection was conducted or, if identified during the latter half of the current 10-year interval, within 4 years after	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			the end of the 10-year interval. These additional inspections conducted during the 4 years following the end of an inspection interval cannot also be credited toward the number of inspections for the following 10-year interval. The number of inspections may be limited by the extent of piping or tanks subject to the observed degradation mechanism. 5. The expansion of sample inspections may be halted in a piping system or portion of system that will be replaced within the 10-year interval during which the inspections were conducted or, if identified during the latter half of the current 10-year interval, within 4 years after the end of the 10-year interval. 6. Indications of cracking are evaluated in accordance with applicable codes and plant-specific design criteria. 7. Unacceptable cathodic protection survey results are entered into the CAP. l) Update the Underground Piping and Tanks Asset Management Plan to include: 1. The required inspections for SLR. 2. A requirement that newly installed, buried stainless steel piping is coated in accordance with Table 1 of National Association of Corrosion Engineers (NACE) SP0169-2007 or Section 3.4 of NACE RP0285-2002. m) Replace the in-scope copper alloy tubing used in the diesel fuel oil system between the ‘A’ diesel fuel oil storage tank and the	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			diesel fire pump with 304 stainless steel tubing encased within a 4" stainless steel chase pipe.	
31	Internal Coatings/Linings for In-scope Piping, Piping Components, Heat Exchangers, and Tanks (A.2.2.28)	XI.M42	Implement the new Internal Coatings/Linings for In-Scope Piping, Piping Components, Heat Exchangers, and Tanks AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and perform pre-SPEO baseline inspections no earlier than 10 years prior to the SPEO.
32	ASME Section XI Subsection IWE (A.2.2.29)	XI.S1	Continue the existing ASME Section XI, Subsection IWE AMP, including enhancements to: a) Specify the preventive actions for storage, lubricants, and SCC potential discussed in Section 2 of Research Council for Structural Connections publication “Specification for Structural Joints Using High-Strength Bolts,” for structural bolting consisting of ASTM A325, ASTM A490, and equivalent bolts. b) Monitor accessible portions of high-temperature (temperatures above 140°F) drywell piping penetrations that are not pressurized during local leak rate testing and have no CLB fatigue analysis to detect cracking.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			c) Perform supplemental one-time surface or enhanced visual examinations comprising a representative sample (five per unit) of the stainless steel penetrations or DMWs associated with high-temperature (temperatures above 140°F) stainless steel piping systems in frequent use to confirm the absence of SCC aging effects. d) Specify a one-time volumetric examination of metal shell surfaces that are inaccessible from one side if triggered by plant-specific OE identified after the date of issuance of the initial renewed license. If triggered, this inspection will be performed by sampling randomly selected, as well as focused, metal shell locations susceptible to corrosion that are inaccessible from one side. The trigger for this one-time examination is plant-specific occurrence or recurrence of metal shell corrosion (base metal material loss exceeding 10 percent of nominal plate thickness) that is determined to originate from the inaccessible side. Guidance provided in EPRI TR–107514 will be considered when establishing a sampling plan. This sampling is conducted to demonstrate, with 95 percent confidence, that 95 percent of the accessible portion of the metal shell is not experiencing greater than 10 percent wall loss. e) If SCC is identified as a result of the supplemental one-time inspections, additional inspections will be conducted in accordance with the site's corrective action process. This will include incrementing sample size during the current outage by one additional penetration at a time from the uninspected population of stainless steel penetrations or DMWs associated with high- temperature (greater than 140°F) stainless steel piping	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			systems in frequent use until cracking is no longer detected. Periodic inspection of subject penetrations with DMWs for cracking will be added to the ASME Section XI, Subsection IWE AMP if necessary, depending on the inspection results.	
33	ASME Section XI Subsection IWF (A.2.2.30)	XI.S3	Continue the existing ASME Section XI, Subsection IWF AMP, including enhancements to: a) Evaluate the acceptability of inaccessible areas (e.g., portions of ASME Class 1, 2, and 3 supports encased in concrete, buried underground, or encapsulated by guard pipe) when conditions are identified in accessible areas that could indicate the presence of, or result in, degradation to such inaccessible areas. b) Specify the preventive actions for storage, lubricants, and SCC potential discussed in Section 2 of Research Council for Structural Connections publication “Specification for Structural Joints Using High-Strength Bolts”, for structural bolting consisting of ASTM A325, ASTM A490, and equivalent bolts. c) Perform and document a one-time supplemental inspection of an additional 5 percent of the sample size specified in Table IWF-2500-1 for Class 1, 2, and 3 piping supports. The additional supports will be selected from the remaining population of IWF piping supports and will include components that are most susceptible to age-related degradation (i.e., based on time in service, aggressive environment, etc.). The one-time inspection will occur within five years prior to entering the SPEO. d) Specify that, for component supports with high-strength bolting greater than one-inch nominal diameter, volumetric examination	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement and start the pre-SPEO one-time supplemental support inspections no earlier than 5 years prior to entering the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			comparable to that of ASME Code, Section XI, Table IWB-2500-1, Examination Category B-G-1 will be performed to detect cracking in addition to the VT-3 examination. A representative sample of bolts will be inspected during the inspection interval prior to the start of the SPEO and in each 10-year period during the SPEO. Identify the population of ASME Class 1, 2, 3, and MC high-strength structural bolting greater than one-inch nominal diameter within the boundaries of IWF-1300 and establish a sample to be 20 percent of the population (for a material / environment combination) up to a maximum of 19 bolts. e) If a component support does not exceed the acceptance standards of IWF-3400 but is repaired to as-new condition, the sample is increased or modified to include another support that is representative of the remaining population of supports that were not repaired. f) Specify that the following conditions are also unacceptable: 1. Loss of material due to corrosion or wear; 2. Debris, dirt, or excessive wear that could prevent or restrict sliding of the sliding surfaces as intended in the design basis of the support; 3. Cracked or sheared bolts, including high-strength bolts, and anchors.	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
34	10 CFR Part 50, Appendix J (A.2.2.31)	XI.S4	Continue the existing 10 CFR Part 50, Appendix J AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
35	Masonry Walls (A.2.2.32)	XI.S5	Continue the existing Masonry Walls AMP, including enhancements to: a) Update the implementing procedure to require inspections for masonry walls to be performed every five years	No later than 6 months prior to the SPEO, or no later than the last refueling outage prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
36	Structures Monitoring (A.2.2.33)	XI.S6	Continue the existing Structures Monitoring AMP, including enhancements to: a) Specify that the following component types are contained in the scope: airlocks, ballistic shields, bird screens, louvers, stairs, ladders, handrails, platforms, sliding surfaces, new fuel storage racks, pit boxes, reactor shield wall, refueling water seal assembly, and the RPV pedestal. b) Specify the preventive actions for storage, lubricant selection, and bolting and coating material selection discussed in Section 2 of the Research Council for Structural Connection publication, “Specification for Structural Joints Using High-Strength Bolts,” will	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and perform pre-SPEO baseline tests and inspections prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			be used for structural bolting consisting of American Society for Testing and Materials (ASTM) A325, ASTM A490, ASTM F1852 and/or ASTM F2280 bolts. c) Clarify that in addition to MoS2 lubricants, “other lubricants containing sulfur” will not be used on high- strength bolting. d) Clarify that in addition to watertight, missile and pressure doors, “doors for shelter and protection” are inspected. e) Include monitoring and trending of leakage volumes and chemistry for signs of concrete or steel reinforcement degradation if active through-wall leakage or groundwater infiltration is identified. Considerations may include engineering evaluation, more frequent inspections, or destructive testing of affected concrete to validate existing concrete properties, including concrete pH levels. f) Include surface staining in the list of alkali-silica reaction (ASR) indications. g) Inspect and monitor the loss of material due to pitting or crevice corrosion and cracking in stainless steel and aluminum components. h) Inspect and monitor the loss of mechanical function for sliding surfaces. i) Excavate, inspect, and monitor below-grade, inaccessible concrete structural elements exposed to aggressive groundwater/soil which will include the performance of a baseline visual inspection prior to the SPEO at a minimum of	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			one location which has experienced aggressive groundwater. The baseline inspection results will be used to conduct a baseline evaluation that will determine the additional actions (if any) that are warranted. j) Perform a baseline evaluation prior to the SPEO of the baseline inspection results for below-grade, inaccessible concrete structural elements exposed to aggressive groundwater/soil. The evaluation will consider the baseline inspection results to determine the additional actions (if any) that are warranted. Quantitative baseline inspection results will be evaluated based on the existing acceptance criteria for concrete identified in the implementing procedure. Additional actions may include: enhanced inspection techniques and/or frequency, destructive testing, and/or focused inspections of representative accessible (leading indicator) concrete structural elements, or if it is determined that accessible areas are not suitable leading indicators for the potential aging effects, excavation and inspection of below grade, inaccessible concrete structural elements exposed to aggressive groundwater/soil. The baseline inspection and evaluation results will set the subsequent inspection requirements and inspection intervals (not to exceed 5 years) for the SPEO. k) Perform periodic evaluation updates of below-grade, inaccessible concrete structural elements exposed to aggressive groundwater/soil (not to exceed 5 years). Updates will be based on OE, periodic inspections, and will consider the opportunistic or focused inspection results during the interval. The periodic evaluation results will update subsequent inspection requirements	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			and inspection intervals (not to exceed 5 years) accounting for the extent of degradation experienced to manage the concrete aging for the SPEO as required which may include: enhanced inspection techniques and/or frequency, destructive testing, and/or focused inspections of representative accessible (leading indicator) concrete structural elements, or if it is determined that accessible areas are not suitable leading indicators for the potential aging effects, excavation and inspection of below grade, inaccessible concrete structural elements exposed to aggressive groundwater/soil. l) Inspections for all structures within the scope of the Structures Monitoring AMP to be performed every five years or the next scheduled refueling outages following the five year interval in normally inaccessible areas. m) Include qualification requirements specified in ACI 349.3R-02 for inspection/evaluation team members and examiners. n) Evaluate the acceptability of inaccessible areas when conditions exist in accessible areas that could indicate the presence of, or result in, degradation of such inaccessible areas. o) Include acceptance criteria for the following components: 1. Polystyrene blowout panels (no hardening, loss of strength, blistering, loss of material and no visible cracking that affects the intended function of the panel). 2. Stainless steel and aluminum components (no loss of material and no visible cracking that affects the intended function of the component).	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			3. Sliding surfaces (no indications of excessive loss of material and no debris or dirt that could restrict or prevent sliding of the surfaces as required by design). 4. Structural sealants (no active leakage or loss of sealing that would affect the intended function of the connected structure or nearby components). p) Inspect and monitor the hardening, loss of strength, loss of material, cracking, and blistering in polystyrene blowout panels.	
37	Inspection of Water-Control Structures Associated with Nuclear Power Plants (A.2.2.34)	XI.S7	Continue the existing Inspection of Water-Control Structures Associated with Nuclear Power Plants AMP, including enhancements to: a) Update the implementing procedure to include stop logs in the scope of the Inspection of Water-Control Structures Associated with Nuclear Power Plants AMP. b) Specify the preventive actions for storage, lubricant selection, and bolting and coating material selection discussed in Section 2 of the Research Council for Structural Connection publication, “Specification for Structural Joints Using High-Strength Bolts,” will be used for structural bolting consisting of American Society for Testing and Materials (ASTM) A325, ASTM A490, ASTM F1852 and/or ASTM F2280 bolts. c) Update the implementing procedure to clarify that in addition to MoS2 lubricants, “other lubricants containing sulfur” will not be used on high-strength bolting.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			d) Update the implementing procedure to require inspections for Water-Controlled Structures to be performed every five years. e) Update the implementing procedure to state that further evaluation of evidence of groundwater infiltration or through-concrete leakage may also include destructive testing of affected concrete to validate existing concrete properties, including concrete pH levels, and that assessments may include analysis of the leakage pH, along with mineral, chloride, sulfate, and iron content in the leakage water if leakage volumes allow. f) Update the implementing procedure to evaluate the acceptability of inaccessible areas when conditions exist in accessible areas that could indicate the presence of, or result in, degradation of such inaccessible areas. g) Update the implementing procedure to include provision for special inspections immediately following the occurrence of significant natural phenomena, such as large floods, earthquakes, hurricanes, tornadoes, and intense local rainfalls. h) Include surface staining in the list of alkali-silica reaction (ASR) indications. i) Include qualification requirements specified in ACI 349.3R-02 for inspection/evaluation team members and examiners. j) Prior to the SPEO, sample raw (river) water for pH, chlorides, and sulfates from intake bay locations. Enter adverse results, which exceed water chemistry criteria, into the corrective action program. Monitor river water chemistry on a frequency not to exceed five	

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			years that accounts for seasonal variations (e.g., quarterly monitoring every fifth year).	
38	Protective Coating Monitoring and Maintenance (A.2.2.35)	XI.S8	Continue the existing Protective Coating Monitoring and Maintenance AMP, including enhancements to: a) Update the implementing procedure to specify that if coating areas cannot be inspected, it will be noted in the inspection documentation with a reason why the inspection could not be conducted. b) Update the implementing procedure to reference C4 of RG 1.54 Rev. 3 for maintenance of Service Level I Coatings.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
39	Electrical Insulation for Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (A.2.2.36)	XI.E1	Continue the existing Electrical Insulation for Electrical Cables and Connections Not subject to 10 CFR 50.49 Environmental Qualification Requirement AMP, including an enhancement to: a) Update the implementing procedure to require visual inspections for accessible electrical cables and connections subjected to an ALE be performed every 10 years.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the 10-year interval inspections prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
40	Electrical Insulation for Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits (A.2.2.37)	XI.E2	Continue the existing Electrical Insulation for Electrical Cables and Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements Used in Instrumentation Circuits AMP, including enhancements to: a) Revise the implementing procedures to include documented periodic review of calibration test results for neutron monitors and radiation monitors within the scope of this program, which are not subject to cable circuit testing. Perform the first periodic review for SLR prior to the SPEO and at least every 10 years thereafter.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the 10-year interval inspections prior to the SPEO.
41	Electrical Insulation for Inaccessible Medium-Voltage Power Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (A.2.2.38)	XI.E3A	Continue the existing Electrical Insulation for Inaccessible Medium-Voltage Power Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements AMP, including enhancements to: a) Update definitions of significant moisture as exposure to moisture that lasts more than 3 days (i.e., long term wetting or submergence over a continuous period) that if left unmanaged, could potentially lead to a loss of intended function. b) Include submarine or other cables designed for continuous wetting or submergence as a one-time inspection and test with additional periodic tests and inspections determined by the one-time test/inspection results as well as industry and plant-specific OE. c) Include inspection of pull boxes equipped with remote water level monitoring and alarms that result in consistent and subsequent pumpout of accumulated water prior to the wetting or submergence of cables at least once every 5 years.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and complete the one-time 5 year and 6 year interval inspections no later than 6 months prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
			d) Inspect pull boxes for water accumulation after event-driven occurrences, such as heavy rain, rapid thawing of ice and snow, or flooding. Pull boxes equipped with remote water level monitoring and alarms are inspected after event-driven occurrences if water accumulation is indicated by the monitoring system. e) Ensure the reliability, self-monitoring features, and operation of continuous remote water level and alarm capabilities of such devices, if installed and credited for 5-year inspection intervals, are demonstrated routinely depending on the attributes of the specific equipment used. f) Test medium-voltage power cables within the scope of this program at least once every 6 years.	
42	Electrical Insulation for Inaccessible Instrument and Control Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (A.2.2.39)	XI.E3B	Implement the new Electrical Insulation for Inaccessible Instrument and Control Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the annual inspections prior to the SPEO.

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
43	Electrical Insulation for Inaccessible Low-Voltage Power Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (A.2.2.40)	XI.E3C	Implement the new Electrical Insulation for Inaccessible Low-Voltage Power Cables Not Subject to 10 CFR 50.49 Environmental Qualification Requirements AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement the AMP and start the 6-year interval inspections prior to the SPEO.
44	Fuse Holders (A.2.2.41)	XI.E5	Implement the new Fuse Holders AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement AMP and start the inspections prior to the SPEO.
45	Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements (A.2.2.42)	XI.E6	Implement the new Electrical Cable Connections Not Subject to 10 CFR 50.49 Environmental Qualification Requirements AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
46	High-Voltage Insulators (A.2.2.43)	XI.E7	Implement the new High-Voltage Insulators AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO. i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037 Implement AMP and start the inspections prior to the SPEO.
47	RHR Heat Exchanger Augmented Inspection Program (A.2.3.1)	N/A	Continue the existing RHR Heat Exchanger Augmented Inspection AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
48	Torus Submerged Components Inspection Program (A.2.3.2)	N/A	Continue the existing Torus Submerged Components Inspection AMP.	No later than the last refueling outage prior to the SPEO, or no later than 6 months prior to the SPEO i.e.: Unit 1: 02/06/2034 Unit 2: 12/13/2037
49	Quality Assurance Program (A.1.3)	Appendix A	The Quality Assurance Program is an existing program that is credited.	Ongoing

Table A-3

List of SLR Implementation Actions and Implementation Schedule

No.	Aging Management Program or Activity (Section)	NUREG-2191 Section	Action Description	Implementation Schedule
50	Operating Experience Program (A.1.4)	Appendix B	The Operating Experience Program is an existing program that is credited.	Ongoing